Observer Effects in Mechanistic Interpretability

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Abstract

Interpretability methods intervene on a model to read out its mechanisms, but the intervention can itself change the model's natural state: an activation forced off the data manifold may drive a pathway that is silent in the ordinary forward pass, fabricating a circuit that is an artifact of measurement rather than a feature of the model. Such off-manifold divergence is pervasive, and prior work shows it can manufacture illusory attributions—yet not all divergence is consequential, and which divergences are *pernicious* rather than *harmless* has remained uncharacterised. We study this on the polysemantic SIIT benchmark, transformers with known, overlapping task circuits where a false positive is well defined, and propose a two-axis mechanistic typology of dormancy—*gate-crossing*, where a component's own patch crosses a nonlinear boundary, and *coupling*, where it displaces the live circuit off-manifold through interaction—with an instrument for each axis: a Curvature Score and an interaction-weighted off-manifold ripple. Across intervention regimes these signals separate pernicious from harmless components precisely where off-manifold magnitude cannot, and the two axes prove complementary. We present this as an exploratory measurement framework: falsifiable mechanistic signatures for when an intervention pollutes the very computation it aims to reveal.

1 Introduction

Mechanistic interpretability probes a model by intervening on it and reading the effect on behaviour, on the premise that the intervention exposes a mechanism already present. Yet the intervention also perturbs the model's state: forced off the manifold of activations the model naturally visits, a representation can drive components silent in the ordinary forward pass, manufacturing an effect that belongs to the intervention rather than the computation—a *dormant circuit*. This measurement-induced artifact has surfaced from two directions: optimised subspaces that encode information the model does not causally use [Makelov et al., 2023], and off-manifold *divergence* in alignment search and activation patching that can awaken a dormant pathway [Grant et al., 2025]. The difficulty both leave open is that *not all divergence is pernicious*—every intervention diverges, and most of it is inert—so holding an intervention on-manifold is neither always possible nor the right target. What is missing is a characterisation of *which* divergences fabricate dormant circuits and which are harmless. Our concern is thus not whether one estimator efficiently approximates another, but a prior question about the intervention itself: when does the tool used to study a model change the very state it means to reveal?

We take up this question where a false positive is well defined—the polysemantic SIIT benchmark [Anders and Garriga-Alonso, 2024, Gupta et al., 2024], whose transformers implement known, overlapping task circuits with off-circuit components provably inert (§2). Our central proposal is a *mechanistic typology of dormancy* along two axes. In *gate-crossing* (self-induced) dormancy a component's own patch crosses a nonlinear boundary and switches on a pathway inert on the natural run; in *coupling* (interaction-induced) dormancy the patch instead displaces the live in-circuit

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components off their manifolds through its interaction with the residual stream. The typology connects two previously separate quantities—the curvature that makes attribution patching unfaithful [Kramár et al., 2024], and the mediator–bypass interaction term—to the concrete mechanisms by which an intervention fabricates a circuit.

We make three contributions. **(i)** The typology, which recasts the pernicious-versus-harmless question as a two-mechanism discrimination and ties curvature and interaction to dormancy. **(ii)** An instrument for each axis—a magnitude-normalised *Curvature Score* and an interaction-weighted *off-manifold ripple*. **(iii)** Evidence on ground truth that, at matched divergence where magnitude alone is exhausted, these instruments separate pernicious from harmless components where geometry cannot; that the two axes are complementary, the ripple catching precisely the components curvature misses; and that a partner-identity signal separates *borrowed* dormancy (recruiting another task's circuit) from *epotic* dormancy (aligned with no task), while the intervention strategy dissociates the mechanisms.

2 Background

Activation patching identifies the components that causally mediate a task [Mueller et al., 2026, Heimersheim and Nanda, 2024] by estimating the natural indirect effect (NIE) of replacing a mediator's clean representation with its counterfactual one. *Noising* holds the bypass—every component but the mediator—at the clean input and corrupts only the mediator; *denoising* holds the bypass in the counterfactual world and restores only the mediator. Because the residual stream lets the bypass read from and write to the mediator, the noising estimate does not isolate the mediator's path-specific effect but also carries its interaction with the clean bypass [Vaidyanathan et al., 2026]. Denoising removes this coupling and measures the pure indirect effect (PIE), so that $\mathrm{NIE} = \mathrm{PIE} + \mathrm{INT}$, the interaction term INT being the second-order cross-curvature of the readout in the mediator and bypass perturbations. Attribution patching [Kramár et al., 2024, Syed et al., 2023] replaces the counterfactual re-run with the first-order Taylor term, an efficient approximation of the NIE.

Establishing which divergence is pernicious requires ground truth, since a false positive is well defined only when a component's true role is known. We work in the polysemantic SIIT benchmark [Anders and Garriga-Alonso, 2024, Gupta et al., 2024]: transformers trained to implement known, overlapping multi-task circuits under the Strict Interchange Intervention Loss, which penalises any off-circuit component that performs non-trivial computation. An off-circuit component thus contributes nothing to the natural run, so when an intervention makes it appear in-circuit the behaviour is attributable to the intervention and the dormant label is sound. A shared MLP overlapping all task circuits acts as a task router [Merullo et al., 2024, Lan et al., 2023], supplying the cross-task interaction geometry we study.

3 A Typology of Dormancy

A *dormant circuit* is a set of components that does not participate in a task's computation during the model's natural run, but is activated by an intervention—leading to the illusory conclusion that it belongs to the task's circuit when it is in fact a measurement artifact. We propose that dormancy arises through two distinct pathways. In *self-induced* dormancy, divergence at the intervention site drives the site's activation off-manifold, inducing behaviour inconsistent with the natural run of the model. In *coupled* dormancy, the intervened activation interacts with the residual stream and causes a ripple in components *away from* the site; the divergence of those interacting components is what produces the confirmatory behaviour that marks the intervened site as part of the circuit. Because every intervention induces *some* off-manifold divergence, magnitude alone cannot separate a pernicious divergence—one that fabricates confirmatory behaviour—from a harmless one (§2). We argue that perniciousness is determined not by how far a representation is pushed off-manifold, but by *which of the two pathways* the divergence engages, and we organise the paper around instruments that detect each one.

3.1 Curvature Induced Dormancy

The nonlinearities of the transformer blocks supply the structure for self-induced dormancy. For a fixed ReLU activation pattern, for example, a component acts as an affine map $f = W_{\mathrm{eff}}x + b_{\mathrm{eff}}$; an intervention that drives a pre-activation across the active/inactive boundary switches the pattern, and

the local gradient of f changes with it. The magnitude of that change is the curvature the intervention traverses, and it is what makes the gradient diagnostic of dormancy. A genuine in-circuit component is already active on the natural run, so an intervention perturbs it within a locally linear regime and its effect is well captured by the local gradient. A dormant component instead sits in the flat, inactive region on the natural run; it can contribute to the output only if the intervention pushes it *across* the boundary into the active region—a crossing the natural-run gradient cannot anticipate. The resulting effect is therefore not a property of the natural computation but an artifact manufactured by the intervention's region-crossing.

Kramár et al. [2024] study this same curvature, but within the question of whether attribution patching (AtP) faithfully approximates activation patching. AtP estimates an intervention's effect from the local gradient at the clean activation, and they show this estimate degrades precisely when an endpoint—the clean or the counterfactual pre-activation—lies in a flat, saturated region of a nonlinearity such as the attention softmax, where the local gradient no longer reflects the true patch. For them the gap between the gradient estimate and the true effect is an approximation *error*, to be corrected (e.g. by recomputing the saturated step explicitly) so that AtP better matches full activation patching. We take up the same geometry with a different question: how can we exploit this curvature to identify an unfaithful intervention?

The magnitude of the divergence alone does not capture this curvature-induced dormancy, because whether an intervention crosses a gate depends on the curvature of the nonlinearity at the component *and* the direction in which the intervention displaces the pre-activation, not on the size of the displacement by itself. Two interventions can push a component equally far off its manifold, yet only the one whose displacement points across a region boundary flips the activation pattern and manufactures a dormant effect; an equal displacement that stays within the same linear region leaves the computation intact. We therefore isolate this region-crossing contribution from the raw off-manifold magnitude—comparing the full patch to its first-order approximation and normalising for effect size—as the *Curvature Score* (§4), which is non-zero exactly when the specific displacement crosses a boundary. While ReLU gives the sharpest such boundary, the effect is generic to any saturating nonlinearity, from the softmax of the attention heads to the GeLU of more modern transformers.

3.2 Interaction Induced Dormancy

A second mechanism produces dormancy through the mediator's interaction with the rest of the circuit via the residual stream. When the mediator is corrupted, this interaction can push the *live, in-circuit* components off *their* manifolds, and it is this induced divergence of the partners that inverts the output and confirms the mediator's apparent importance. The coupling is captured by the interaction term of the causal mediation decomposition, $\mathrm{INT} = \mathrm{NIE} - \mathrm{PIE}$ (§2): a large INT with negligible PIE is the signature of a component whose effect is carried almost entirely by its interaction with the bypass.

The interaction magnitude alone, however, cannot distinguish *pernicious* coupled dormancy from *healthy* context-dependent components. Such a component contributes little to restoring the clean output when the rest of the residual stream is corrupted, giving a low PIE; yet it is part of the ground-truth circuit, so corrupting it in the noising run still breaks the output and gives a high NIE. It therefore presents the same high-INT, low-PIE signature as pernicious coupled dormancy, and discriminating the two requires the *type* of interaction rather than its size. We therefore resolve INT into its pairwise, partner-specific contributions (xINT), which report *which* components a single patch of the mediator interacts with, and combine this with the off-manifold divergence of those partners—an off-manifold *ripple* that measures how far the mediator, once patched, displaces its high-interaction in-circuit partners off their manifolds (§4). Because the polysemantic benchmark labels every site, the partner identities additionally separate two sub-types of coupled dormancy: *borrowing*, where the mediator recruits another task's components, and *epotic* dormancy, where it interacts with components belonging to no task's circuit.

Together these give a two-axis typology (Fig. 1): a divergence is pernicious if it either crosses a gate (Axis 1, curvature) or couples to the live circuit (Axis 2, interaction), and harmless if it does neither. The two mechanisms are not mutually exclusive—a single component may both cross a nonlinear boundary and, through its interaction with the bypass, push the live circuit off-manifold—so a dormant attribution may arise from either mechanism or from both at once, and the instruments for the two axes

are complementary rather than competing. The typology thus recasts the pernicious-versus-harmless question as a two-mechanism discrimination problem, and §4–§5 show that instruments targeting each axis separate pernicious from harmless divergence where geometric magnitude cannot.

4 Instruments

We employ a range of interchange methods in order to extract intervened representations that span different geometries and interaction regimes, and to fit predictors that classify dormancy from their geometric and dynamic characteristics. We first describe the patching strategies, then the geometric divergence metrics, and finally the two mechanistic instruments that target the axes of §3: the curvature score and the off-manifold ripple for gate-crossing and interaction induced dormancy respectively.

4.1 Activation patching

Our two base strategies are symmetric token replacement and Gaussian patching [Zhang and Nanda, 2024]:

Definition 1. Symmetric Token Replacement (STR) is an interchange intervention that defines the base (clean) and source (counterfactual) inputs by running an interchange exchange (INTINV) on the causal variables (V) of the clean representation (h_b), the tokens causally relevant for the output,with replacements from semantically close substitutes from the counterfactual (corrupted, h_s) run.

$$INTINV(M, b, V) \leftarrow INTINV(M^L, s, V)h_b[V] \leftarrow h_s[V] \tag{1}$$

Definition 2. Gaussian patching perturbs the activation at a site with scaled isotropic noise,

$$h_b \;\leftarrow\; h_b + \alpha \cdot 3\,\sigma_{h_b}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I). \tag{2}$$

Push variants. We additionally define STR-push and Gaussian-push methods that apply the intervention at the MLP shared across the task circuits, in the polysemantic benchmark models, and read out its effect at downstream components:

$$\text{STR push:}\quad h_X \leftarrow h_X + \alpha\,(h_s - h_b), \qquad \text{Gaussian push:}\quad h_X \leftarrow h_X + \alpha \cdot 3\,\sigma_{h_X}\,\varepsilon, \tag{3}$$

where X is the shared MLP site, h_s and h_b are the source and base representations, and α doses the intervention, with $\alpha = 1$ recovering a full swap or Gaussian intervention at X. These methods serve two purposes. First, they diversify the intervened representations available for fitting predictors beyond what the swap strategies alone provide. More importantly, they let us reproduce, in a controlled benchmark, the cross-task interaction geometry that is generic to real transformers. In a real model essentially every component is polysemantic, it participates in several tasks at once, so an ordinary patch of any single component perturbs multiple task computations simultaneously and induces cross-task interactions in its downstream effects. Our benchmark concentrates this polysemanticity in a single shared MLP. By pushing that MLP and reading out its effect at downstream components, we deliberately induce the same downstream consequence of perturbing a component that serves several tasks and thereby approximate what an ordinary single-component patch would produce in a fully polysemantic model.

A further variation, applied to both the swap and push forms of STR, draws the base and source from *different* tasks. This cross-task setting simulates worst-case divergent, dormancy-prone interventions on which to stress-test our predictors, and lets us study curvature-induced dormancy in the regime where the divergence points along a direction that activates *another* task's circuit; the within-task setting (base and source drawn from the same task, as in a standard STR swap) instead probes the realistic near-manifold regime.

Noising and denoising under a push. *Noising* imposes on c its push-induced representation h'_c, its activation when X is pushed, while the bypass, including X, is left clean, yielding $\mathrm{NIE}(c)$. *Denoising* injects each components state under the intervention on the MLP, so the bypass (including the pushed MLP) is in the corrupted world r', while restoring only c to its clean representation h_c, yielding $\mathrm{PIE}(c)$. The interaction $\mathrm{INT}(c) = \mathrm{NIE}(c) - \mathrm{PIE}(c)$ is non-degenerate because the bypass differs between the two cells; unlike the self-patch, the push supplies a genuine corrupted-bypass world.

Gaussian patching applied at the measured component itself, by contrast, carries *only* geometric signal. Because the only corrupted variable is the mediator, the corrupted-bypass world coincides with the clean-bypass world; there is no valid denoising run, the pure indirect effect equals the natural indirect effect, and $\mathrm{INT} \equiv 0$ by construction (§2). Gaussian self-patching therefore serves as an interaction-free baseline that isolates off-manifold divergence from any coupling signal.

4.2 Divergence metrics

We construct a per-site activation manifold from the singular value decomposition of the clean representations across examples,

$$h = U\Sigma Q^\top, \qquad h \in \mathbb{R}^{N \times d_{\text{model}}}, \tag{4}$$

retaining the top 8 right singular vectors Q_s—the principal directions of variation—as a low-dimensional manifold [Fefferman et al., 2016] that captures the dominant variation without uninformative dimensions. Writing $\delta = h'_c - h_c$ for the intervened divergence at a site, we measure its two geometric components against these manifolds,

$$\mathrm{orth}_B(\delta) = \big\|(I - Q_B Q_B^\top)\,\delta\big\|, \qquad \mathrm{along}_S(\delta) = \big\|Q_S^\top \delta\big\|, \tag{5}$$

where orth_B is the component perpendicular to the base-task manifold Q_B (the task whose circuit we trace) and along_S is the component lying within a target-task subspace Q_S. For STR cross-task the target is the source task; otherwise we take along_S against each non-base task. As shown in §3, the magnitude of this divergence is not itself decisive; the remaining instruments target the two mechanisms that turn an off-manifold divergence from harmless to pernicious.

4.3 The Curvature Score

To operationalise the curvature-induced dormancy axis (§3) we compare the true effect of a noising patch to its first-order attribution. Let Ψ denote the downstream map from the activation at site c to the output logits, and let the *readout* Φ be the scalar summary of that output we attribute, the logit difference toward the source, $\Phi = \mathrm{logit}[\mathrm{src}] - \mathrm{logit}[\mathrm{base}]$, so that Φ depends on h_c through Ψ. The noising effect of patching c is the induced change in this readout, $\mathrm{NIE}(c) = \Phi(h_c + \delta_c) - \Phi(h_c)$, and attribution patching approximates it by the first-order term $\widehat{\mathrm{NIE}}_{\mathrm{lin}}(c) = \langle \delta_c,\, \nabla_{h_c}\Phi|_{\text{clean}} \rangle$, substituting the clean-point gradient for a re-run.

Expanding the readout to second order about the clean activation gives

$$\Phi(h_c + \delta_c) - \Phi(h_c) = \langle \delta_c,\, \nabla_{h_c}\Phi \rangle + \tfrac{1}{2}\,\delta_c^\top H\,\delta_c + \mathcal{O}(\|\delta_c\|^3), \qquad H = \nabla^2_{h_c}\Phi\big|_{\text{clean}}, \tag{6}$$

where H is the Hessian—the curvature of the readout at the clean activation. Subtracting the first-order term leaves the curvature contribution, $\mathrm{NIE}(c) - \widehat{\mathrm{NIE}}_{\mathrm{lin}}(c) \approx \frac{1}{2}\,\delta_c^\top H\,\delta_c$, an approximation that is accurate whenever the cubic and higher terms are negligible [Zhang and Wang, 2026]. We define the *Curvature Score* as this residual, normalised by the raw effect size,

$$\kappa_c \;=\; \frac{\big|\,\mathrm{NIE}(c) - \widehat{\mathrm{NIE}}_{\mathrm{lin}}(c)\,\big|}{\overline{|\Phi_n - Y_a|}} \;\approx\; \frac{\big|\tfrac{1}{2}\,\delta_c^\top H\,\delta_c\big|}{\overline{|\Phi_n - Y_a|}}, \tag{7}$$

with Φ_n and Y_a the noising and clean readouts and the denominator, the mean absolute noising effect, normalising for raw effect size.

This residual is exactly what vanishes when no boundary is crossed. For a fixed ReLU activation pattern the readout is affine in h_c (§3), so $H = 0$ and the first-order term is *exact*; the curvature term becomes non-zero only when δ_c carries the pre-activation across a boundary into a different affine region, where the effective linear map—and hence the local slope—changes. A large κ therefore signals that the effect was manufactured by a region-crossing rather than captured by the natural-run gradient, while the normalisation removes the raw off-manifold magnitude. For the smooth nonlinearities of the attention softmax and GeLU, H is the ordinary Hessian and is largest in the curved transition regions, giving the same signal.

4.4 Partner-resolved interaction and the off-manifold ripple

The interaction term $\mathrm{INT} = \mathrm{NIE} - \mathrm{PIE}$ (§2) measures the size of a component's coupling to the bypass but not its *type*. To recover the type we decompose the interaction across partners.

Definition 3 (Multi-component interaction)**.** Patching a set C of components jointly, the sub-graph effect decomposes as

$$\mathrm{NIE}(C) = \sum_{M \in C} \mathrm{NIE}(M) + \sum_{T \subseteq C,\, |T| \geq 2} \mathrm{xINT}(T), \tag{8}$$

where $\mathrm{xINT}(T)$ is the cross-interaction among the members of T.

We restrict to pairwise interactions ($|C| = 2$): the SIIT models are small enough that we take higher-order terms to be largely subsumed by the pairwise signal. For a pair $C = \{c, p\}$ the joint patch recovers partner p's contribution to $\mathrm{INT}(c)$,

$$\mathrm{xINT}(c, p) = \mathrm{NIE}(\{c, p\}) - \mathrm{NIE}(c) - \mathrm{NIE}(p), \tag{9}$$

so $\mathrm{xINT}(c, \cdot)$ profiles c's magnitude of interaction with each component in the bypass. Because the polysemantic benchmark labels every site, we split this profile into interaction with on-base-task, other-task, and non-aligned components, enabling a predictor that distinguishes coupled dormancy which *borrows* another task's components from *epotic* dormancy of a component in no task's circuit.

Finally, to convert interaction type into a dormancy signal we combine xINT with the divergence metric. If a mediator has deceptively high NIE but is in fact dormant, one mechanism is that patching it pushes the in-circuit components of the clean bypass off *their* manifolds, inverting the output. We therefore estimate the off-manifold displacement that a single patch of c induces on its high-xINT in-circuit partners. Writing $\delta_{N|c} = h_{N|c} - h_N$ for the change induced at partner N when c is imposed on an otherwise clean bypass, the ripple is the $|\mathrm{xINT}|$-weighted mean of the partners' off-manifold displacement over the in-circuit set $\mathcal{C}$,

$$\texttt{ripple_xint_in}(c) = \frac{\sum_{N \in \mathcal{C}} |\mathrm{xINT}(c, N)|\, \mathrm{orth}_B(\delta_{N|c})}{\sum_{N \in \mathcal{C}} |\mathrm{xINT}(c, N)|}. \tag{10}$$

The same construction over an arbitrary partner set $\mathcal{P}$ defines the general weighted ripple

$$\texttt{ripple_xint}(c; \mathcal{P}) = \frac{\sum_{N \in \mathcal{P}} |\mathrm{xINT}(c, N)|\, \mathrm{orth}_B(\delta_{N|c})}{\sum_{N \in \mathcal{P}} |\mathrm{xINT}(c, N)|}, \tag{11}$$

of which `ripple_xint_in` is the restriction to the in-circuit partners $\mathcal{C}$, while taking $\mathcal{P}$ over all partners gives the grid-wide `ripple_xint`. The in-circuit restriction is our primary signal, since coupled dormancy operates by displacing the *live* circuit off-manifold.

4.5 Evaluation

On the polysemantic SIIT circuits every site carries a ground-truth label, so a false positive is well defined. A *pernicious* component is an off-circuit site whose attributed |NIE| reaches that of a genuine circuit member (the per-combination median), and a *harmless* component is any other off-circuit site. The benchmark's labels further split the pernicious components into two sub-types: a *borrowed* false positive is off the base-task circuit but is a member of *another* task's circuit, so the intervention recruits a genuine component of a different task, whereas an *epotic* false positive belongs to no task's circuit and is a purely non-aligned artifact. Per intervention regime we fit logistic separations of pernicious from harmless components, reporting each instrument's AUC, its gain over the orth baseline, and its *high-orth* AUC—the separation restricted to already-divergent components, where orth is exhausted and only a mechanistic signal can discriminate. All estimates are cluster-bootstrapped over task combinations.

5 Findings

5.1 Mechanistic Instruments resolve Pernicious vs. Harmless Divergence.

Geometric magnitude (orth) is necessary but not sufficient for perniciousness: many highly-divergent components are harmless, so the decisive test is at *matched* divergence—the high-orth regime where orth are exhausted and can no longer separate dormant from non-dormant. There the Axis-1 instrument, the Curvature Score (κ), recovers the discrimination: on within-task swaps it separates the

Table 1: False-positive vs. harmless separation, standalone AUC (* 95% CI excludes 0.5). Final column: hi-orth AUC of the dominant detector (Curvature for swaps, along_{nb} for pushes)—separation persists among already-divergent components.

Regime	orth	Curvature	along_{nb}	$\text{along}_{\text{home}}$	xINT	$\text{ripple}^{\text{in}}_{\text{xINT}}$	Curv. hi-orth
STR swap (cross)	0.81^*	0.90^*	0.87^*	0.81^*	0.56	0.60^*	0.86^*
STR swap (within)	0.61^*	0.86^*	0.81^*	0.69	0.49	0.55	0.87^*
STR push (cross)	0.88^*	0.69^*	0.91^*	0.97^*	0.51	0.58	0.84^*
STR push (within)	0.84^*	0.73^*	0.89^*	0.98^*	0.63^*	0.53	0.83^*
Gaussian push	0.93^*	0.88^*	0.94^*	–	0.79^*	0.54	0.87^*
Gaussian self-push	0.80^β	–	–	–	$\equiv 0$	–	–

$^\beta$ standardised slope of $|\text{NIE}| \sim \text{orth}$ (self-push has no FP separation; INT$\equiv 0$). along_{nb} = `along_nonbase_max`; $\text{along}_{\text{home}}$ is AUC on the borrowed subset (a component that is a member of a non-base task's circuit). ripple AUC on the high-divergence subset where defined.

Table 2: Independent effect: standardised logistic coefficients β from a joint model per regime on the *full* off-circuit FP set (* 95% CI excludes 0). Curvature carries within-task swaps (orth is not distinguishable from 0); the directional detector leads pushes; INT contributes independently only within-task. $\text{ripple}^{\text{in}}_{\text{xINT}}$ (‡) is estimated on the subset where it is defined (the ripple-candidate set).

Regime	orth	Curvature	along_{nb}	xINT	$\text{ripple}^{\text{in},\ddagger}_{\text{xINT}}$
STR swap (cross)	$+1.02^*$	$+1.89^*$	$+0.80^*$	$+0.19^*$	$+1.05^*$
STR swap (within)	$+0.05$	$+1.65^*$	$+0.51^*$	-0.08	$+0.64^*$
STR push (cross)	$+0.52^*$	$+0.22$	$+1.36^*$	$+0.11$	$+0.15$
STR push (within)	$+0.72^*$	$+0.56^*$	$+1.30^*$	$+0.33^*$	-0.07
Gaussian push	$+1.12^*$	$+0.75^*$	$+0.60$	$+0.30$	-0.28
Gaussian self-push	$+0.82^{*\,\dagger}$	–	–	$\equiv 0$	–

† $|\text{NIE}| \sim \text{orth}$ slope (geometric null). ‡ ripple on the ripple-candidate (orth-gated) subset, the only rows where it is defined; all other columns on the full off-circuit FP set. The ripple *block* adds for both push strata (Table 3).

two at AUC 0.86, and at 0.87 among the already-divergent (hi-orth) components (Table 1). In the joint model its standardised effect ($\beta = +1.65$) dwarfs that of orth, whose independent contribution is indistinguishable from zero ($+0.05$, n.s., Table 2): once divergence is held high, geometric magnitude contributes essentially nothing and the gate-crossing signal carries the discrimination. The same holds on cross-task swaps (curvature AUC 0.90, hi-orth 0.86; $\beta = +1.89$).

Both instruments are non-redundant predictors, and together they separate pernicious from harmless divergence in the regime where off-manifold magnitude alone cannot. In the joint model on the STR swap regime, the Curvature Score and the off-manifold ripple (`ripple_xint_in`) are *each* significant ($\beta = +1.89$ and $+1.05$ cross-task; $+1.65$ and $+0.64$ within-task, Table 2). Neither is a proxy for the other, so the two respond to distinct components rather than to a shared signal. The residual analysis shows that they operate on distinct regions of dormant components, as the components that the orth + Curvature Score model mis-classifies, the ripple metric recovers the pernicious ones in every STR regime (residual AUC 0.77–0.81 on cross and within-task swaps and cross-task push Table 3). The two instruments therefore partition the pernicious population into a curvature-detected part and a ripple-detected part in curvature's blind spot, each operating on a separate region of the divergence. It is these instruments, not the magnitude of the off-manifold divergence, that discriminate pernicious from harmless.

5.2 Strategy dissociation: gate-crossing versus directional recruitment

The intervention strategy determines which pathway a dormant divergence engages. A symmetric token replacement substitutes a representation, a discrete jump that can carry a component across a region boundary and manufacture a gate-crossing effect. A push at the shared MLP instead slides activations smoothly along a task direction, steering downstream components into another task's subspace [Xiong et al., 2026, Wurgaft et al., 2026] without traversing any boundary. The two strategies therefore surface different dormancy: swaps foreground the curvature and interaction induced dormancy discussed in §3, whereas the push produces a distinct directional recruitment

Table 3: Ripple metrics in the Curvature Score's blind spot: AUC for separating false positives from harmless components *among those the* $\mathrm{orth} + \kappa$ *model mis-classifies* (* 95% CI excludes 0.5). The ripple block adds in all four STR regimes but is muted within-task; Gaussian shows no routed contagion.

Ripple metric	STR swap (cross)	STR swap (within)	STR push (cross)	STR push (within)	Gaussian push
`ripple_in_orth`	0.58	0.49	0.64	0.47	0.52
`ripple_off_orth`	0.40	0.49	0.51	0.46	0.68
`ripple_frac`	0.67	0.60	0.63^*	0.60	0.48
`ripple_xint_orth`	0.62	0.59	0.60	0.56	0.63
`ripple_xint_in_orth`	0.64^*	0.56	0.62	0.49	0.48
Ripple block (5)	0.78^*	0.81^*	0.77^*	0.58^*	0.74
κ-error FPs (n)	78	93	77	101	42

Table 4: The directional signal is independent of curvature under the push. Standardised logistic coefficient of along before and after adding the curvature score, per regime (* 95% interval excludes 0). Curvature absorbs much of the directional signal on the swaps but almost none under the push.

Regime	β_{along} (no curvature)	β_{along} (+ curvature)	change
STR swap (cross)	$+1.11^*$	$+0.79^*$	-29%
STR swap (within)	$+0.84^*$	$+0.51^*$	-39%
STR push (cross)	$+1.33^*$	$+1.33^*$	0%
STR push (within)	$+1.38^*$	$+1.22^*$	-12%

that neither the curvature score nor the interaction instruments detect. We characterise the push regime here and show that its perniciousness is carried by the *direction* of the divergence, not by its off-manifold magnitude or its curvature.

The shared MLP acts as a task router. Pushing the shared MLP toward a task's subspace recruits downstream components into that subspace, and this directed recruitment is what fabricates the confirmatory behaviour. Alignment with a component's own home-task subspace separates the borrowed false positives almost perfectly, and alignment with any non-base subspace separates the epotic ones, both reported per regime alongside the other detectors in Table 1. The recruitment is targeted rather than incidental: borrowing is enriched among false positives (35 to 43% against a 25 to 28% base rate), and under a cross-task push 72% of borrowed false positives are home to the pushed target task. The recruited representation lands inside another task's data. Writing $d_K(h'_c)$ for the whitened distance of the intervened representation h'_c to the data manifold of the non-base task K that its divergence most aligns with, normalised so that a task's own representations have median distance 1, borrowed false positives sit at $d_K \approx 3$, well within the ≈ 8.6 of an unrelated representation (Table 5). The push routes members into another task's data rather than off the manifold.

The directional signal is independent of curvature rather than a proxy for it, as confirmed by controlling for the curvature score. Under the swaps the standardised coefficient of along is substantially reduced once the Curvature Score enters the model (a 29 to 39% drop), since there the two are correlated and curvature carries much of the shared signal. Under the push the same control leaves along essentially unchanged (Table 4): its recruitment signal is orthogonal to curvature, not a reflection of it.

Borrowed and epotic recruitment are geometrically opposite. The two false-positive sub-types differ in where the recruitment lands, and the divergence metrics alone mislead about this; only the intervened representation resolves it. Both move with comparable magnitude into a task direction ($\mathrm{along}_K \approx 2.5$ to 2.8), but the borrowed divergence also escapes its home subspace ($\mathrm{orth}_{\mathrm{home}} \approx 2.9$) while the epotic divergence stays tangent to the task it points at ($\mathrm{orth}_K \approx 0.5$). Read on the divergence, this suggests the borrowed component leaves the manifold and the epotic one remains on it. The intervened representation reverses that reading: borrowed representations land near their home task's data (whitened distance $d_K \approx 3$, below the ≈ 8.6 of an unrelated representation), recruited into a genuine region, whereas epotic representations land as far from the task they align with as an unrelated representation ($d_K \approx 11$), aligned with a direction but never entering its data (Table 5). The

Table 5: Borrowed versus epotic false positives under the push (medians). The divergence metrics on δ_c ($\mathrm{orth}_{\text{base}}$, along_K, orth_K) suggest the borrowed component leaves the manifold and the epotic one stays on it; the representation distance d_K to the aligned non-base task's data shows the reverse. Reference column: an unrelated representation scored against the same task.

Sub-type	$\mathrm{orth}_{\text{base}}$	along_K	orth_K	d_K	reference
Borrowed	1.7	2.8	2.9	**3**	8.6
Epotic	0.3	2.5	0.5	**11**	8.6

distinction follows from the reference point of the two measures. Every orth and along quantity is computed on the divergence δ_c against a task's tangent subspace, whereas d_K compares the resulting representation to the task's data density, which is why it is decisive.

The push carries no curvature, and off-manifold magnitude separates without being the mechanism. Because the push slides smoothly rather than crossing a boundary, it produces almost no curvature for the curvature score to read. The raw region-crossing residual on false positives is ≈ 0.05 under the cross push and ≈ 0.15 under the within push, against ≈ 1.5 to 3.0 for the swaps, while harmless components carry essentially zero in every regime. Push dormancy does not proceed by gate-crossing, so there is little for a second-order instrument to detect; the off-manifold ripple, also a second-order quantity, is weak for the same reason. The off-manifold magnitude orth retains statistical separability under the push (AUC 0.84 to 0.88, Table 1), because harmless push components diverge even less, yet its absolute value is small (median 0.43). Perniciousness in this regime is set by the direction of the divergence, not its magnitude: along is the dominant separator (AUC 0.89 to 0.91), while orth and the Curvature Score each separate only through a weak by-product of that recruitment. orth responds to the small off-manifold residue the directional push leaves behind, and the Curvature Score to the negligible curvature of its near-linear response; neither reflects the recruitment itself.

6 Limitations and Future Work

The main limitation is that we evaluate on semi-synthetic SIIT transformers rather than large pretrained models, so neither the dormancy types nor the instruments have yet been validated at scale. We chose this testbed deliberately. Dormancy is defined relative to a component's true role, and deciding whether an attribution is a false positive requires ground-truth circuit labels, which only a synthetic benchmark provides; establishing dormancy without that ground truth is precisely the difficulty our setting removes. The clearest threat to transfer is neuron-level polysemanticity. We have not verified with sparse autoencoders that the benchmark's neurons carry multiple features, and if they are monosemantic these models would differ from real transformers, where polysemanticity is pervasive. We address this in part by concentrating multi-task computation in a shared MLP and probing it with the push interventions, which reproduce at the component level what a patch of a polysemantic unit does: perturbing a component that serves several tasks and inducing cross-task effects downstream. Because these components are genuinely cross-task, we expect them to be neuron-polysemantic as well. More fundamentally, the mechanisms we identify, gate-crossing and interaction, are properties of a component's local nonlinearity and its coupling to the residual stream, so we expect them to hold at the circuit level regardless of neuron-level semantics. Confirming this on larger models, with features recovered by sparse autoencoders standing in for the ground-truth circuits, is the natural next step.

A second limitation is operational. The directional metrics `along_home` and `along_nb`, on which the push regime's directed recruitment is best detected, require the subspaces of the other tasks a component can participate in. These are known on the benchmark but not in a real model, where the full set of a component's cross-task manifolds is unavailable, so these detectors are not directly deployable. The curvature score and the interaction ripple carry no such dependence, but they do not fire on the directional-recruitment regime. A label-free detector for that regime, for example an unsupervised density estimate of the intervened representation against data-driven rather than ground-truth subspaces, remains open.

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